



Preliminary Data Sheet of NHP8000VT3

NORYL™ NHP8000VT3 resin is a modified polyphenylene ether (PPE) for extrusion and injection molding. This grade contains non-brominated, non-chlorinated flame retardant and carries a UL94 flame rating of V0 at $\geq$ 0.25mm and a Comparative Tracking Index (CTI) rating of PLC0 by ASTM D3638.

TYPICAL PROPERTIES	NHP8000VT3	UNIT	STANDARD
MECHANICAL			
Tensile Modulus, 5mm/min	2491	MPa	ASTM D638
Tensile Strength, yield, 5mm/min	66.8	MPa	ASTM D638
Tensile Strain, break, 5mm/min	10.8	%	ASTM D638
Tensile Strain, yield, 5mm/min	4.2	%	ASTM D638
Flexural Modulus	2600	MPa	ASTM D790
Flexural strength, 5% strain	108	MPa	ASTM D790
Izod Impact, notched, 23 °C	211	J/m	ASTM D256
Izod Impact, notched, -40 °C	84	J/m	ASTM D256
Izod Impact, un-notched, 23 °C	No break	J/m	ASTM D4812
Izod Impact, un-notched, -40 °C	1830	J/m	ASTM D4812
THERMAL			
HDT, 0.45 MPa, 3.2 mm	150	°C	ASTM D648
HDT, 1.82 MPa, 3.2 mm	133	°C	ASTM D648
Vicat Softening Temp, 10N/50°C/h	166	°C	ASTM D1525
Vicat Softening Temp, 50N/50°C/h	155	°C	ASTM D1525
BPT	140	°C	IEC Standard
PHYSICAL			
MFR (300°C, 5.0 kg)	11.7	g/10min	ASTM D 1238
Density	1.1	g/cm ³	ISO 1183



Shrinkage, flow	0.73	%	Reference ASTM D955
Shrinkage, x-flow	0.79	%	Reference ASTM D955
Water absorption, 23°C/24hrs	0.04	%	ISO 62
ELECTRICAL			
Surface Resistivity	7.50E+16	Ω /square	ASTM D257
Volume Resistivity	6.20E+15	Ω -cm	ASTM D257
Comparative Tracking Index	PLC0*		ASTM D3638
FLAME			
VX, 0.25mm	V-0		UL94
VX, 0.3mm	V-0		UL94
VX, 0.5mm	V-0		UL94
VX, 0.75mm	V-0		UL94
VX, 1.0mm	V-0		UL94
5VA	2.0mm		UL94
5VB	2.0mm		UL94
LOI	46.4	%	ISO 4589

*Pass from low voltage (300V) to high voltage (600V).



Injection Molding / Extrusion Guidelines

Parameter Title	Minimum	Maximum	Units
Drying Temperature	100	120	°C
Drying Time	3	5	h
Melt Temperature	280	330	°C
Nozzle Temperature	250	330	°C
Front – Zone 3 Temperature	280	330	°C
Middle – Zone 2 Temperature	280	330	°C
Rear – Zone 1 Temperature	280	330	°C
Mold Temperature	90	120	°C
Back Pressure	50	80	bar
Holding Pressure	1000	1700	bar
Cooling Time	30	50	s
Screw Speed	60	100	rpm

1) Typical values only. Variations within normal tolerances are possible for various colors. All properties, except the MFR rate are measured on injection molded samples.

2) Only typical data for material selection purpose. Not to be used for part or tool design.

3) This rating is not intended to reflect hazards presented by this or any other material under actual fire conditions.

4) Own measurement according to UL.

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